

Python - Access Tuple Items

Access Tuple Items

The most common way to access values within a Python tuple is using indexing, We just need to specify the index of the elements we want to retrieve to the square bracket `[]` notation.

In Python, a tuple is an immutable ordered collection of elements. "Immutable" means that once a tuple is created, we cannot modify or change its contents. We can use tuples to group together related data elements, similar to lists, but with the key difference that tuples are immutable, while lists are mutable.

In addition to indexing, Python provides various other ways to access tuple items such as slicing, negative indexing, extracting a subtuple from a tuple etc. Let us go through this one-by-one –

Accessing Tuple Items with Indexing

Each element in a tuple corresponds to an index. The index starts from 0 for the first element and increments by one for each subsequent element. Index of the last item in the tuple is always "length-1", where "length" represents the total number of items in the tuple. To access the elements of a tuple we just need to specify the index of the item we need to access/retrieve, as shown below –

```
tuple[3]
```

Example

Following is the basic example to access tuple items with slicing index –

```
tuple1 = ("Rohan", "Physics", 21, 69.75)
tuple2 = (1, 2, 3, 4, 5)

print ("Item at 0th index in tuple1: ", tuple1[0])
print ("Item at index 2 in tuple2: ", tuple2[2])
```

It will produce the following output –

```
Item at 0th index in tuple1: Rohan
Item at index 2 in tuple2: 3
```

Accessing Tuple Items with Negative Indexing

Negative indexing in Python is used to access elements from the end of a tuple, with **-1** referring to the last element, **-2** to the second last, and so on.

We can also access tuple items with negative indexing by using negative integers to represent positions from the end of the tuple.

Example

In the following example, we are accessing tuple items with negative indexing –

```
tup1 = ("a", "b", "c", "d")
tup2 = (25.50, True, -55, 1+2j)

print ("Item at 0th index in tup1: ", tup1[-1])
print ("Item at index 2 in tup2: ", tup2[-3])
```

We get the output as shown below –

```
Item at 0th index in tup1: d
Item at index 2 in tup2: True
```

Accessing Range of Tuple Items with Negative Indexing

By range of tuple items, we mean accessing a subset of elements from a tuple using slicing. Therefore, we can access a range of tuple items with negative indexing by using the slicing operation in Python.

Example

In the example below, we are accessing a range of tuple items by using negative indexing –

```
tup1 = ("a", "b", "c", "d")
tup2 = (1, 2, 3, 4, 5)

print ("Items from index 1 to last in tup1: ", tup1[1:])
print ("Items from index 2 to last in tup2", tup2[2:-1])
```

It will produce the following output –

```
Items from index 1 to last in tup1: ('b', 'c', 'd')
Items from index 2 to last in tup2: (3, 4)
```

Access Tuple Items with Slice Operator

The slice operator in Python is used to fetch one or more items from the tuple. We can access tuple items with the slice operator by specifying the range of indices we want to extract. It uses the following syntax –

```
[start:stop]
```

Where,

- **start** is the starting index (inclusive).
- **stop** is the ending index (exclusive).

Example

In the following example, we are retrieving subtuple from index 1 to last in "tuple1" and index 0 to 1 in "tuple2", and retrieving all elements in "tuple3" –

```
tuple1 = ("a", "b", "c", "d")
tuple2 = (25.50, True, -55, 1+2j)
tuple3 = (1, 2, 3, 4, 5)
tuple4 = ("Rohan", "Physics", 21, 69.75)

print ("Items from index 1 to last in tuple1: ", tuple1[1:])
print ("Items from index 0 to 1 in tuple2: ", tuple2[:2])
print ("Items from index 0 to index last in tuple3", tuple3[:])
```

Following is the output of the above code –

```
Items from index 1 to last in tuple1: ('b', 'c', 'd')
Items from index 0 to 1 in tuple2: (25.5, True)
Items from index 0 to index last in tuple3 ('Rohan', 'Physics', 21, 69.75)
```

Accessing Sub Tuple from a Tuple

A subtuple is a part of a tuple that consists of a consecutive sequence of elements from the original tuple.

We can access a subtuple from a tuple by using the slice operator with appropriate start and stop indices. It uses the following syntax –

```
my_tuple[start:stop]
```

Where,

- **start** is the starting index (inclusive).
- **stop** is the ending index (exclusive) of the subtuple.

If we does not provide any indices, the slice operator defaults to starting from index 0 and stopping at the last item in the tuple.

Example

In this example, we are fetching subtuple from index "1 to 2" in "tuple1" and index "0 to 1" in "tuple2" using slice operator –

```
tuple1 = ("a", "b", "c", "d")
tuple2 = (25.50, True, -55, 1+2j)

print ("Items from index 1 to 2 in tuple1: ", tuple1[1:3])
print ("Items from index 0 to 1 in tuple2: ", tuple2[0:2])
```

The output obtained is as follows –

```
Items from index 1 to 2 in tuple1: ('b', 'c')
Items from index 0 to 1 in tuple2: (25.5, True)
```

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